Supervised Classification of Face Mask Images using Logistic Regression and SVM

Irina Alexandra Ciurescu

Practical Machine Learning – Project 1

1 Introduction

The goal of this project is to study a simple image classification problem using supervised machine learning methods. The task is to classify face images into two classes: people wearing a face mask and people not wearing a face mask. The dataset used is the Face Mask Detection 12K Images Dataset, publicly available on Kaggle [1].

Two classification models were tested: Logistic Regression and Support Vector Machine (SVM). Since the models cannot work directly with raw images, two different feature representations were used: Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT) combined with Bag of Visual Words (BoVW). The main goal of the experiments is to compare the performance of the two models and to see how the choice of feature representation affects the classification results.

2 Dataset and Preprocessing

2.1 Dataset

The dataset used in this project is the Face Mask 12K Image Dataset. It contains images divided into two classes: *WithMask* and *WithoutMask*. The dataset is already split into training, validation and test sets. The number of images in each set is shown in Table 1.

Table 1: Number of images in each dataset split

Split	**WithMask**	**WithoutMask**	**Total**
Train	5000	5000	10000
Validation	400	400	800
Test	483	509	992

2.2 Preprocessing

The images in the dataset have different sizes, so some preprocessing was necessary before extracting the features.

For the HOG representation, all images were converted to grayscale and resized to 64×64 pixels. Using the same image size ensured that all HOG feature vectors have the same length.

For SIFT, the images were also converted to grayscale, but they were kept at their original size. SIFT detects local keypoints in each image, so resizing was not required for this representation.

The training, validation and test sets were kept separate during all experiments. The test set was only used for final testing, not for choosing preprocessing settings or model hyperparameters.

3 Feature Representations

3.1 HOG

Histogram of Oriented Gradients (HOG) describes an image using the directions of intensity changes, or gradients. This allows HOG to capture information about edges and shapes present in the image. For this representation, each image was converted to grayscale and resized to 64×64 pixels. HOG features were then extracted using 9 orientations, cells of 8×8 pixels and blocks of 2×2 cells. L2-Hys normalization was used for the blocks. With these settings, each image was represented by a feature vector containing 1764 values.

3.2 SIFT and Bag of Visual Words

SIFT detects distinctive local points in an image and creates a descriptor for each keypoint. Unlike HOG, the number of SIFT descriptors can be different for each image. Thus, the SIFT descriptors cannot be used directly as fixed-sized input for the classification models.

That is why SIFT was combined with the Bag of Visual Words method. First, the SIFT descriptors were extracted from the training images. Then, a random sample of 100,000 descriptors was used to build a visual vocabulary with K-Means. The vocabulary contained 100 visual words.

For each image, its SIFT descriptors were assigned to the closest visual words. A histogram was then created by counting how many descriptors belonged to each visual word. The histogram was normalized, resulting in a feature vector with 100 values for every image.

4 Models and Experimental Setup

Two supervised classification models were used: Logistic Regression and Support Vector Machine (SVM). Both models were tested using the HOG and SIFT/BoVW feature representations.

Before training, the feature values were standardized using `StandardScaler`. The scaler was fitted only on the training data and then applied to the validation and test sets.

4.1 Logistic Regression

The main hyperparameters tested for Logistic Regression were the regularization parameter C and the type of penalty. The following values were used:

- $C \in \{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$
- penalty: L1 and L2

The `liblinear` solver was used because it supports both L1 and L2 regularization.

4.2 Support Vector Machine

The second model used was SVM. Two kernel types were tested: a linear kernel and the Radial Basis Function (RBF) kernel. For the linear kernel, different values of C were tested:

$$C \in \{0.01, 0.1, 1, 10\}$$

For the RBF kernel, both C and γ were varied:

$$\gamma \in \{\text{scale}, 0.0001, 0.001\}$$

Each configuration was trained on the training set and evaluated using validation accuracy. The model with the highest validation accuracy was selected separately for HOG and SIFT/BoVW features.

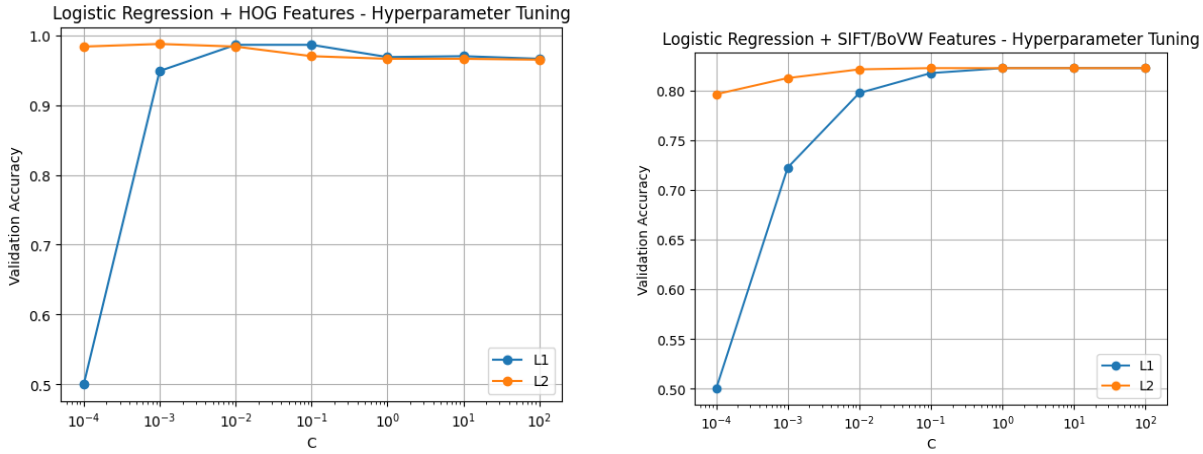
5 Results

This section presents the results obtained during hyperparameter tuning and the final evaluation on the test set.

5.1 Hyperparameter Tuning

5.1.1 Logistic Regression

For Logistic Regression, different values of C were tested with both L1 and L2 regularization for the two feature representations.



(a) HOG features

(b) SIFT/BoVW features

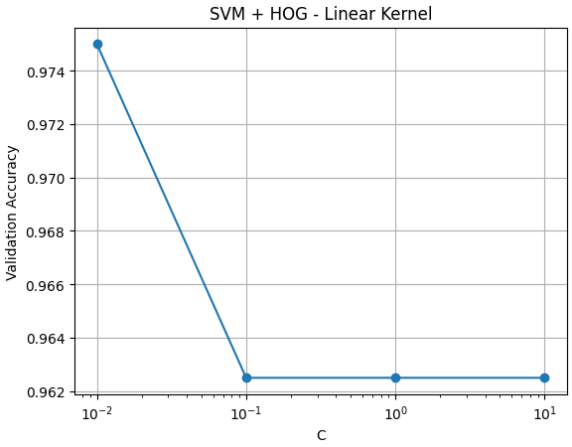
Figure 1: Validation accuracy of Logistic Regression for different hyperparameter values.

For HOG features, the best configuration was obtained with $C = \mathbf{0.001}$ and **L2** regularization, reaching a validation accuracy of **0.9875**.

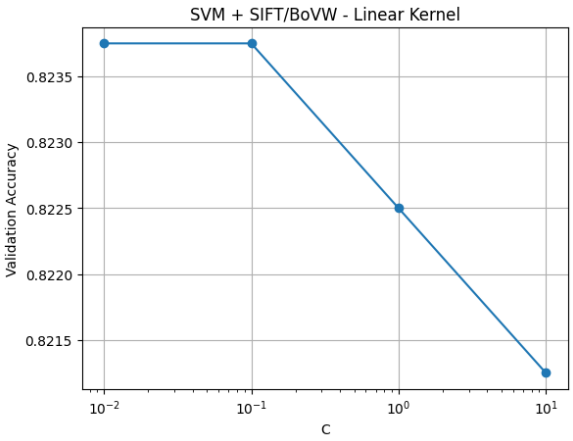
For SIFT/BoVW features, the best configuration used $C = \mathbf{1}$ and **L1** regularization, with a validation accuracy of **0.8225**.

5.1.2 Support Vector Machine

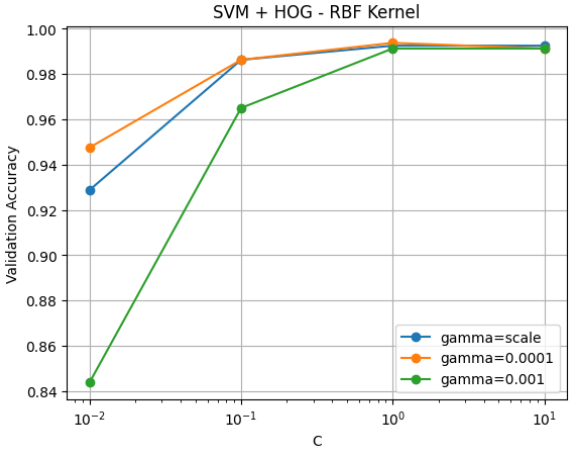
For SVM, both linear and RBF kernels were tested. Different values of C were evaluated for the linear kernel, while both C and γ were varied for the RBF kernel.



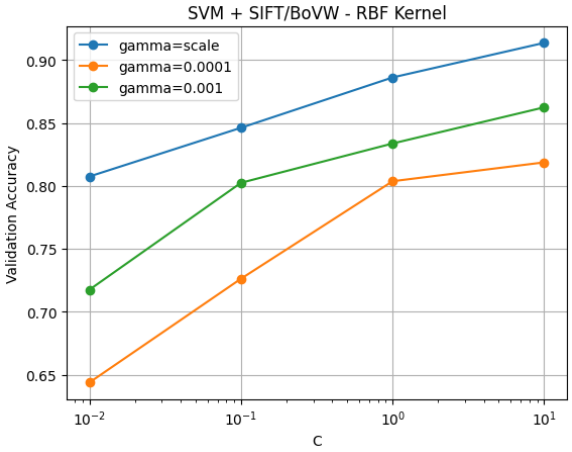
(a) Linear SVM – HOG



(b) Linear SVM – SIFT/BoVW



(c) RBF SVM – HOG



(d) RBF SVM – SIFT/BoVW

Figure 2: Validation accuracy of SVM for different hyperparameter values.

For HOG features, the best configuration used the RBF kernel with $C = 1$ and $\gamma = 0.0001$, obtaining a validation accuracy of 0.99375.

For SIFT/BoVW features, the best configuration also used the RBF kernel, with $C = 10$ and $\gamma = \text{scale}$, obtaining a validation accuracy of 0.91375. However, in this case the difference between training and validation accuracy is quite large, around 8.6 percentage points, which suggests some overfitting. This is different from the HOG results, where the difference is only about 0.7 percentage points.

Table 2 summarizes the best configuration found for each model and feature representation.

Table 2: Best configurations based on validation accuracy.

Model	**Features**	**Best Parameters**	**Val. Accuracy**
Logistic Regression	HOG	$C = 0.001$, L2	0.9875
Logistic Regression	SIFT/BoVW	$C = 1$, L1	0.8225
SVM	HOG	$C = 1$, RBF, $\gamma = 0.0001$	0.99375
SVM	SIFT/BoVW	$C = 10$, RBF, $\gamma = \text{scale}$	0.91375

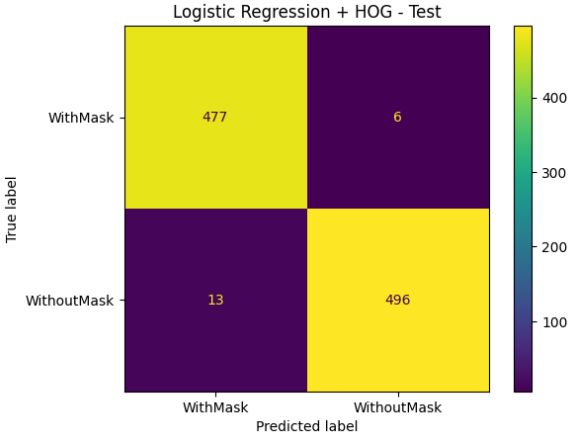
5.2 Final Test Results

After selecting the best hyperparameters using the validation set, the final models were evaluated on the test set.

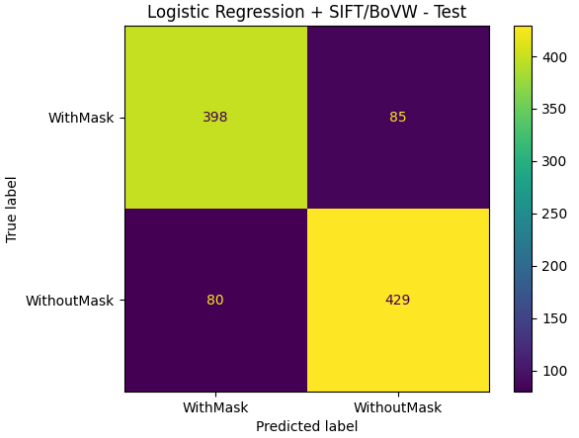
Table 3 shows the test accuracy obtained by each model and feature representation.

Table 3: Final accuracy on the test set.

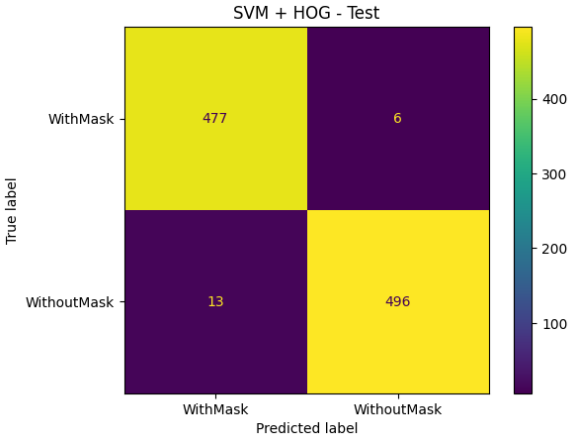
Model	Features	Test Accuracy
Logistic Regression	HOG	0.98085
Logistic Regression	SIFT/BoVW	0.83367
SVM	HOG	0.98085
SVM	SIFT/BoVW	0.92641



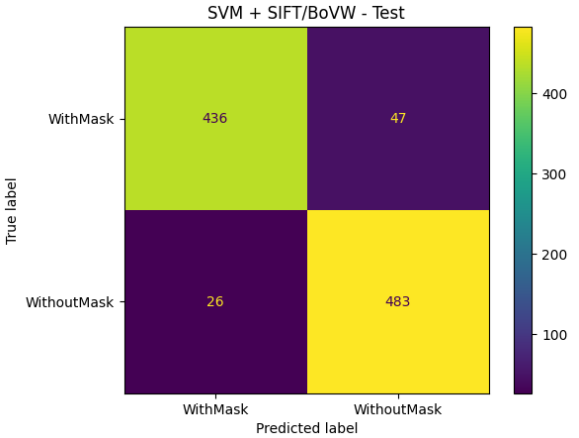
(a) Logistic Regression – HOG



(b) Logistic Regression – SIFT/BoVW



(c) SVM – HOG



(d) SVM – SIFT/BoVW

Figure 3: Confusion matrices for the final models on the test set.

The best test results were obtained using HOG features. Logistic Regression and SVM achieved the same test accuracy of 0.98085 and, in this case, also produced the same predictions, with only 19 misclassified images.

For SIFT/BoVW features, SVM performed considerably better than Logistic Regression, reaching an accuracy of 0.92641 compared to 0.83367. Overall, the test results confirm that HOG was the stronger feature representation for this classification task.

6 Comparison with Existing Work

A recent study using the same Face Mask 12K dataset compared several classical machine learning methods, including Logistic Regression and SVM, with deep learning models such as CNN, MobileNetV2 and ResNet50, and stacked ensembles combining them. For the FM12k dataset, Logistic Regression and SVM achieved accuracies of approximately 0.90–0.91, while the best stacked ensemble reached an accuracy of 0.995 [2].

In this project, the best test accuracy was 0.98085, obtained by both Logistic Regression and SVM using HOG features. However, the experimental settings are different. The referenced study used a stratified 70%/15%/15% train-validation-test split and resized the images to 128×128 pixels, while this project used the original dataset splits and handcrafted HOG and SIFT/BoVW feature representations. Therefore, the results give a useful point of comparison, but they should not be considered directly equivalent.

7 Discussion

The experiments suggest that the feature representation had an important role in the final performance. HOG produced more consistent results across both classifiers, while SIFT/BoVW was more sensitive to the choice of model.

The SIFT/BoVW representation also has some limitations in this experiment. A fixed vocabulary size of 100 visual words was used, built from a random sample of 100,000 training descriptors. Other vocabulary sizes or a larger descriptor sample could have been tested and might lead to different results.

8 Conclusion

This project compared Logistic Regression and SVM for face mask classification using two different feature representations: HOG and SIFT/BoVW. The experiments showed that HOG was the more effective representation for this task, while SIFT/BoVW produced weaker and less consistent results.

Overall, the best final performance was obtained with HOG features, for which both Logistic Regression and SVM reached the same test accuracy.

References

[1] A. Jangra, *Face Mask Detection ∼12K Images Dataset*, Kaggle, accessed September 2026. Available online: `https://www.kaggle.com/datasets/ashishjangra27/face-mask-12k-images-dataset`

[2] R. Kumari, P. Pallavi and P. Saurabh, *Enhancing face mask detection performance using stacked neural ensembles in mask fusion*, Discover Artificial Intelligence, vol. 6, article 171, 2026. doi: 10.1007/s44163-025-00826-4.